



Traditional method (left) versus drone application (right)

increase the demand for fairly-produced, preservative-free and sustainable food. This change will impact the global demand and security. In India, farmers are neither equipped with latest technologies nor are they trained to adopt these fast. Digital farming is slowly entering Indian farms to assist farmers in better decision-making.

Digital technology holds tremendous potential to transform the Indian agricultural economy and impact the lives of Indian farmers. Major challenges confronting Indian agriculture include unsustainable usage of resources, declining farm productivity, rapidly growing demand for high-quality and safe food, stagnating farm incomes and fragmented land holdings. These can be overcome through sustainable and scalable deployment of digital technologies and infrastructures.

Digitalisation of agri-business, which involves leveraging massive amounts of data from agricultural equipment, soil, weather, seeds and chemicals, using modern computer science to automate agriculture, is playing a key role in meeting global food demand. Indian ag-tech startups—nearly 500 of them—in the hunt for sustainable farming solutions for a rapidly-rising population are changing the face of agriculture.

With the global smart farming market set to grow at a CAGR of 20.27 per cent to reach US\$ 25 billion by 2021, India is at the threshold as it is home to one-fifth of the global small-hold farmers. Numerous farm-level applications of digital technology, including remote sensing, geographic information system (GIS), crop and soil health monitoring, and livestock and farm management, have been making their mark.

At the pre-harvest level, digital technology can recommend crop and

input selection, and assist in obtaining credit and insurance, weather advisories, and disease- and pest-related assistance through data generation. Advanced analytics allow farmers to make smart decisions about farming and benefit from economical use of inputs and labour.

However, farmers are always challenged to grow more while coping with volatile weather conditions to meet the rising demand for food of higher quality. This demand for improving efficiency on inputs and outputs, and various technologies has led to improved efficiency. The Indian agri-business sector stands to greatly benefit from the numerous innovative digital technologies across farm, post farm, processing, market and logistics stages that have emerged in the recent past. This article highlights some of the key digital technologies that can truly transform India's agri-business sector.

Biosciences. To increase yields and reduce costs, farmers are using a wide variety of new bioscience technologies and optimising inputs for agricultural production. A Bengaluru-based startup has developed a product in the biostimulant segment to provide growers with a reliable solution to improve crop quality and productivity. Providing inputs to farmers on crop and soil management during the cropping cycle has led to remarkable results in Madhya Pradesh, achieving increased yields of forty per cent for garlic.

Farm optimisation and monitoring services. The IoT, Big Data, cloud and mobility are increasingly being leveraged to improve farm output. A number of service providers on the farming-as-a-service (FaaS) model are leveraging digital technology to provide innovative farm-to-fork solutions to farmers and agri-businesses.

Precision agriculture. It is being adopted by farmers to optimise their use of resources, be it water or fertilisers, to bring efficiency to the farms in terms of cost, yield and more. Further, precision agriculture supports decision-making in irrigation, crop management, water optimisation and the like.

Precision agriculture technologies, ranging from robots, sensors, drones, satellite imagery, Big Data and the IoT, among others, are being employed in various stages of agriculture, from environment analysis, irrigated landscape mapping, soil analysis to crop health analysis. These technologies bring in reliable quality data that help in timely crop damage assessment, crop planning, diversification and contingency planning. Given below are a few examples of technologies being used in precision agriculture.

- Weather forecasting and monitoring is being applied to provide high levels of accuracy in short, medium and long terms.

- Numerous innovations with solar-powered phase-change-enabled materials are being used for irrigation and cold storage, with the aim of catering to small-holder farms and regions with limited or no electricity.

- Eco-friendly crop-protection methods have the potential to minimise a significant proportion of the damage caused by pests and diseases without overdosing crops and plants with chemicals, thus preventing soil and water contamination.

- Sensors that can provide farmers with information about crop yields, rainfall, pest infestation and soil nutrition are being used. These are invaluable for production and offer precise data that can be used to improve farming techniques over time.

- The IoT is being applied to sensors that can detect soil moisture, crop growth and livestock feed levels, to remotely manage and control smart connected harvesters and irrigation equipment. It is also being applied to utilise artificial intelligence (AI)-based analytics to quickly analyse operational data combined with such third-party information as weather information to provide new insights. This is helping in decision-making in terms of crop selection, crop outputs, etc.